conjunction-screen-sgp4	Conjunction screening of TLE catalogues with SGP4, KD-tree coarse pass and closed-form TCA refinement	PDF
star-albers	Albers equal-area conic projection	PDF
star-anomaly	Mean, eccentric, hyperbolic and true anomaly	PDF
star-atmosphere	U.S. Standard Atmosphere 1976 (below 86 km)	PDF
star-audit	Geodetic conversion accuracy audit	PDF
star-calendar	Calendar dates and Julian day numbers, exactly	PDF
star-cdm	Strict CCSDS 508.0-B-1 Conjunction Data Message (KVN) parser with consistency report	PDF
star-chebyshev	Chebyshev series with their derivative	PDF
star-coords	Spherical and cylindrical ⇄ rectangular coordinates	PDF
star-coverage	What a satellite sees, and from where it is seen	PDF
star-crc	Cyclic redundancy checks, bit by bit	PDF
star-crosscheck	Cross-solver verification fabric (S.T.A.R.)	PDF
star-cw	Relative motion near a circular orbit (Clohessy-Wiltshire)	PDF
star-doppler	Range, range rate and one-way Doppler shift	PDF
star-ecliptic	Mean obliquity and equatorial ⇄ ecliptic directions	PDF
star-elements	Classical orbital elements to and from Cartesian state (Curtis Algorithms 4.2 / 4.5)	PDF
star-ellipsoid	The WGS-84 ellipsoid as a surface	PDF

`star-epoch`	Julian and Besselian epochs	PDF
`star-era`	Earth Rotation Angle and IAU 2006 sidereal time	PDF
`star-euler`	Euler angles for the twelve sequences, one convention	PDF
`star-galactic`	Equatorial ⇄ galactic directions	PDF
`star-gasdynamics`	Isentropic flow and normal shock of a perfect gas	PDF
`star-geodesic`	Distance and azimuth on the WGS-84 ellipsoid	PDF
`star-geodesy`	WGS-84 geodetic to/from ECEF with exact inversion at any altitude	PDF
`star-horizon`	Hour angle and declination ⇄ azimuth and elevation	PDF
`star-interp`	Lagrange interpolation through tabulated points	PDF
`star-iod`	Velocity from three positions (Gibbs, Herrick-Gibbs)	PDF
`star-j2`	What the Earth's oblateness does to an orbit, on average	PDF
`star-kepler`	Two-body propagation with universal variables	PDF
`star-lambert`	Universal-variable Lambert orbital boundary value problem solver	PDF
`star-lcc`	Lambert conformal conic projection	PDF
`star-linefit`	Least-squares line and correlation, correctly rounded	PDF
`star-linsolve`	Small linear systems, correctly rounded	PDF
`star-maneuver`	Closed-form vis-viva and Hohmann orbital transfer engine	PDF

star-mat3	The 3×3 matrix operations of frame and attitude code	PDF
star-mercator	Mercator on the ellipsoid, and Web Mercator	PDF
star-moon	Where the Moon is, to about 0.4 degree (0.2 degree for 95 % of dates)	PDF
star-mrp	Modified Rodrigues parameters	PDF
star-orbit	Numerical orbit propagation (Cowell, fixed-step RK4) with two-body + J2 gravity, linked to star_timescales	PDF
star-polar	Polar stereographic projection and UPS	PDF
star-poly	Real roots of quadratics and cubics	PDF
star-polyfit	Least-squares polynomial, correctly rounded	PDF
star-precession	Between J2000 and the mean equator of date (IAU 1976)	PDF
star-quadrature	Gauss-Legendre quadrature	PDF
star-quantile	Median, quantiles, IQR and MAD, correctly rounded	PDF
star-quaternion	Rotations with one stated convention	PDF
star-rootfind	Bracketed roots to the last float	PDF
star-rotframe	Position and velocity between an inertial and a rotating frame	PDF
star-sexagesimal	Degrees ⇄ degrees/minutes/seconds and hours/minutes/seconds	PDF
star-sidereal	Greenwich mean sidereal time (IAU 1982)	PDF

`star-sigma`	What "n sigma" means in one, two and three dimensions	PDF
`star-slerp`	Interpolation of attitudes along the shortest arc	PDF
`star-spacepacket`	CCSDS Space Packets, built and read strictly	PDF
`star-sphere`	Angles between directions on the sphere	PDF
`star-spline`	Natural cubic spline, correctly rounded	PDF
`star-stats`	Summary statistics, correctly rounded	PDF
`star-sun`	Sun direction, beta angle and eclipse test	PDF
`star-tabint`	Integrals of tabulated data, correctly rounded	PDF
`star-tdb`	TDB - TT with the USNO Circular 179 series, verified against ERFA full series within 10 microseconds	PDF
`star-telemetry`	CCSDS AOS/TM transfer frame deframer and Space Packet reassembler	PDF
`star-aos-c`	CCSDS AOS/TM transfer-frame check in C99 (S.T.A.R.)	PDF
`star-timecode`	CCSDS ASCII time codes A and B, strict	PDF
`star-tle`	A strict reader of two-line element sets	PDF
`star-topo`	Look angles from a ground site	PDF
`star-twobody`	The sizes and speeds of a Keplerian ellipse	PDF
`star-units`	Unit conversion with exact definitions and a dimension check	PDF
`star-utm`	Transverse Mercator and UTM coordinates	PDF

`star-vec3`	The three-dimensional vector operations of orbit and attitude code	PDF
`star-wahba`	Attitude from vector observations	PDF
`star-wrap`	Reducing, differencing and averaging angles	PDF